

Relations and Functions

- **Exercise 1**

Let

$$A = \{1, 2, 3\}.$$

Consider the relation

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}.$$

Determine whether R is:

1. reflexive ✓.
2. symmetric ✓.
3. antisymmetric ×.
4. transitive ✓.

- **Exercise 2**

Determine which of the following relations are equivalence relations.

1. On $\mathbb{Z}$,

$$x \sim y \iff x - y \text{ is even.} \quad \checkmark$$

2. On $\mathbb{R}$,

$$x \sim y \iff x \leq y. \quad \times$$

3. On $\mathbb{R}$,

$$x \sim y \iff |x| = |y|. \quad \checkmark$$

4. On $\mathbb{Z}$,

$$x \sim y \iff |x - y| \leq 1. \quad \times$$

- **Exercise 3**

Decide whether each of the following relations is a partial order.

1. On $\mathbb{R}$,

$$x \sim y \iff x \leq y. \quad \checkmark$$

2. On $\mathbb{R}$,

$$x \sim y \iff x < y. \quad \times$$

3. On the power set $\mathcal{P}(X)$,

$$A \sim B \iff A \subseteq B. \quad \checkmark$$

4. On the positive integers,

$$a \sim b \iff a \mid b. \quad \checkmark$$

- **Exercise 4**

Let

$$A = \{1, 2, 3\}, \quad B = \{a, b, c\}, \quad X = A \cup B.$$

Consider the following relations on X . Each relation is specified by listing all ordered pairs (x, y) for which $x \sim y$.

Which of these relations define a function

$$f: X \rightarrow B.$$

1. $R_1 = \{(1, a), (2, b), (3, c)\}.$ ✓
 2. $R_2 = \{(1, a), (1, b), (2, c), (3, a)\}.$ ×
 3. $R_3 = \{(1, a), (2, b)\}.$ ×
 4. $R_4 = \{(1, a), (2, a), (3, a)\}.$ ✓
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• **Exercise 5**

Let

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

Determine

1. The domain of f

$\mathbb{R}.$

2. The codomain of f

$\mathbb{R}.$

3. The image of f

$[0, \infty).$

Is the image of a function always equal to its codomain? No.

• **Exercise 6**

Consider

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2,$$

and

$$g: \mathbb{R} \rightarrow [0, \infty), \quad g(x) = x^2.$$

Are f and g equal as functions? No.

Now consider

$$h: \mathbb{R} \rightarrow \mathbb{R}, \quad h(x) = |x|^2.$$

Are f and h equal as functions? Yes.

• **Exercise 7**

Properties of a Relation

Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$$

Determine whether R is:

- (a) reflexive;
- (b) symmetric;
- (c) antisymmetric;
- (d) transitive.

◦ **Solution 7.1**

- **(a) Reflexive:** Yes. By definition, a relation R on A is reflexive if for every $x \in A$, $(x, x) \in R$. For $A = \{1, 2, 3, 4\}$, the pairs $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$ are all explicitly present in R .
- **(b) Symmetric:** Yes. By definition, a relation is symmetric if for any $x, y \in A$, $(x, y) \in R \implies (y, x) \in R$. The only distinct pairs in R are $(1, 2)$ and $(2, 1)$. Since $(1, 2) \in R \implies (2, 1) \in R$ and $(2, 1) \in R \implies (1, 2) \in R$, the condition holds.
- **(c) Antisymmetric:** No. By definition, a relation is antisymmetric if for any $x, y \in A$, $((x, y) \in R \wedge (y, x) \in R) \implies x = y$. In this relation, we have $(1, 2) \in R$ and $(2, 1) \in R$, but $1 \neq 2$.
- **(d) Transitive:** Yes. By definition, a relation is transitive if for any $x, y, z \in A$, $((x, y) \in R \wedge (y, z) \in R) \implies (x, z) \in R$. Checking the non-trivial combinations:

$$((1, 2) \in R \wedge (2, 1) \in R) \implies (1, 1) \in R$$

$$((2, 1) \in R \wedge (1, 2) \in R) \implies (2, 2) \in R$$

Any other compositions with diagonal relations automatically satisfy the implication.

• **Exercise 8**

Relations on Integers

For each of the following relations on $\mathbb{Z}$, determine whether it is reflexive, symmetric, antisymmetric, and transitive. For every property that fails, give a counterexample.

(a)

$$a \sim b \iff a = b$$

(b)

$$a \sim b \iff a \leq b$$

(c)

$$a \sim b \iff |a - b| \leq 2$$

◦ **Solution 8.1**

Part (a):

- **Reflexive:** Yes. For any integer $a \in \mathbb{Z}$, the statement $a = a$ is universally true. Therefore, $a \sim a$ holds for all $a \in \mathbb{Z}$.
- **Symmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$. By definition, this means $a = b$. Since equality is symmetric, it follows that $b = a$, which means $b \sim a$.

- **Antisymmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$ and $b \sim a$. By definition, this implies $a = b$ and $b = a$. This directly satisfies the requirement that $a = b$.
- **Transitive:** Yes. Let $a, b, c \in \mathbb{Z}$ such that $a \sim b$ and $b \sim c$. By definition, this yields $a = b$ and $b = c$. Substituting b gives $a = c$, which means $a \sim c$.

◦ **Solution 8.2**

Part (b):

- **Reflexive:** Yes. For any integer $a \in \mathbb{Z}$, the statement $a \leq a$ is always true. Thus, $a \sim a$ holds for all $a \in \mathbb{Z}$.
- **Symmetric:** No. Let $a = 1$ and $b = 2$. We have $1 \leq 2$, so $1 \sim 2$ is true. However, $2 \leq 1$ is false, so $2 \sim 1$ does not hold.
- **Antisymmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$ and $b \sim a$. By definition, this means $a \leq b$ and $b \leq a$. By the properties of the standard order relation on integers, $a \leq b \wedge b \leq a \implies a = b$.
- **Transitive:** Yes. Let $a, b, c \in \mathbb{Z}$ such that $a \sim b$ and $b \sim c$. By definition, this yields $a \leq b$ and $b \leq c$. By the transitivity property of inequalities, $a \leq b \wedge b \leq c \implies a \leq c$, which means $a \sim c$.

◦ **Solution 8.3**

Part (c):

- **Reflexive:** Yes. For any integer $a \in \mathbb{Z}$, we have $|a - a| = |0| = 0$. Since $0 \leq 2$, the condition $|a - a| \leq 2$ is satisfied, so $a \sim a$ holds for all $a \in \mathbb{Z}$.
- **Symmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$. By definition, $|a - b| \leq 2$. Since $|a - b| = |-(b - a)| = |b - a|$, we have $|b - a| \leq 2$, which means $b \sim a$.
- **Antisymmetric:** No. Let $a = 1$ and $b = 2$. We have $|1 - 2| = |-1| = 1 \leq 2$, so $1 \sim 2$ holds. Similarly, $|2 - 1| = |1| = 1 \leq 2$, so $2 \sim 1$ holds. However, $1 \neq 2$.
- **Transitive:** No. Let $a = 1$, $b = 3$, and $c = 5$. We have $|1 - 3| = |-2| = 2 \leq 2$, so $1 \sim 3$ holds. We also have $|3 - 5| = |-2| = 2 \leq 2$, so $3 \sim 5$ holds. However, for a and c , we have $|1 - 5| = |-4| = 4 > 2$, which means $1 \sim 5$ does not hold.

• **Exercise 9**

An Equivalence Relation on Integers
Define a relation $\sim$ on $\mathbb{Z}$ by

$$a \sim b \iff a - b \text{ is divisible by } 4.$$

- (a) Prove that $\sim$ is an equivalence relation.
 (b) Find $[0]$, $[1]$, $[2]$, $[3]$, $[5]$.
 (c) How many distinct equivalence classes are there?

◦ **Solution 9.1**

Part (a):

- **Reflexive:** Let $a \in \mathbb{Z}$. Then $a - a = 0$. Since $0 = 4 \times 0$, 0 is divisible by 4. Thus, $a \sim a$ for all $a \in \mathbb{Z}$.
- **Symmetric:** Let $a, b \in \mathbb{Z}$ such that $a \sim b$. By definition, $a - b = 4k$ for some $k \in \mathbb{Z}$. Then $b - a = -(a - b) = -4k = 4(-k)$. Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 4, so $b \sim a$.
- **Transitive:** Let $a, b, c \in \mathbb{Z}$ such that $a \sim b$ and $b \sim c$. By definition, $a - b = 4k$ and $b - c = 4m$ for some $k, m \in \mathbb{Z}$. Adding these equations gives $(a - b) + (b - c) = 4k + 4m \implies a - c = 4(k + m)$. Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 4, so $a \sim c$.
Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation.

◦ **Solution 9.2**

Part (b):

By definition, $[a] = \{x \in \mathbb{Z} \mid x - a = 4k, k \in \mathbb{Z}\} = \{a + 4k \mid k \in \mathbb{Z}\}$.

- $[0] = \{\dots, -8, -4, 0, 4, 8, \dots\} = \{4k \mid k \in \mathbb{Z}\}$
- $[1] = \{\dots, -7, -3, 1, 5, 9, \dots\} = \{1 + 4k \mid k \in \mathbb{Z}\}$
- $[2] = \{\dots, -6, -2, 2, 6, 10, \dots\} = \{2 + 4k \mid k \in \mathbb{Z}\}$
- $[3] = \{\dots, -5, -1, 3, 7, 11, \dots\} = \{3 + 4k \mid k \in \mathbb{Z}\}$
- $[5] = \{\dots, -3, 1, 5, 9, 13, \dots\} = [1]$

◦ **Solution 9.3****Part (c):**

There are exactly 4 distinct equivalence classes, which are $[0]$, $[1]$, $[2]$, and $[3]$.

• **Exercise 10**

Same Remainder

Let

$$A = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

Define a relation $\sim$ on A by

$$a \sim b \iff a \text{ and } b \text{ have the same remainder when divided by 3.}$$

(a) Write down all equivalence classes.

(b) Write the relation $\sim$ explicitly as a set of ordered pairs.

◦ **Solution 10.1****Part (a):**

We group the elements of A by their remainder when divided by 3:

- Remainder 0: $[0] = \{0, 3, 6\}$
- Remainder 1: $[1] = \{1, 4, 7\}$
- Remainder 2: $[2] = \{2, 5\}$

◦ **Solution 10.2****Part (b):**

The relation $\sim$ consists of all pairs (x, y) such that x and y belong to the same equivalence class:

$$\begin{aligned} \sim = \{ & (0, 0), (0, 3), (0, 6), (3, 0), (3, 3), (3, 6), (6, 0), (6, 3), (6, 6), \\ & (1, 1), (1, 4), (1, 7), (4, 1), (4, 4), (4, 7), (7, 1), (7, 4), (7, 7), \\ & (2, 2), (2, 5), (5, 2), (5, 5) \} \end{aligned}$$

• **Exercise 11**

A Partial Order on Sets

Let

$$X = \{1, 2, 3\}.$$

Consider the relation $\subseteq$ on $\mathcal{P}(X)$. Prove that $\subseteq$ is a partial order.

◦ **Solution 11.1****Proof:**

- **Reflexive:** Let $A \in \mathcal{P}(X)$. For any $x \in A$, the statement $x \in A$ is true, so $A \subseteq A$ holds by definition of subset.
- **Antisymmetric:** Let $A, B \in \mathcal{P}(X)$ such that $A \subseteq B$ and $B \subseteq A$. By definition of subset,

$A \subseteq B \implies (\forall x \in A \implies x \in B)$ and $B \subseteq A \implies (\forall x \in B \implies x \in A)$. Thus $x \in A \iff x \in B$, which implies $A = B$.

- **Transitive:** Let $A, B, C \in \mathcal{P}(X)$ such that $A \subseteq B$ and $B \subseteq C$. Let $x \in A$. Since $A \subseteq B$, it follows that $x \in B$. Since $B \subseteq C$ and $x \in B$, it follows that $x \in C$. Thus $x \in A \implies x \in C$, meaning $A \subseteq C$.

Since $\subseteq$ is reflexive, antisymmetric, and transitive, it is a partial order.

• Exercise 12

Divisibility as a Partial Order

Let $\mathbb{N}$ denote the set of positive integers. Define a relation $\preceq$ on $\mathbb{N}$ by

$$x \preceq y \iff x \mid y.$$

Recall that $x \mid y$ means that there exists $k \in \mathbb{N}$ such that $y = xk$. Prove that $\preceq$ is a partial order on $\mathbb{N}$.

◦ Solution 12.1

Proof:

- **Reflexive:** Let $x \in \mathbb{N}$. We can write $x = x \times 1$. Since $1 \in \mathbb{N}$, there exists a positive integer $k = 1$ such that $x = xk$, meaning $x \mid x$. Thus, $x \preceq x$.
- **Antisymmetric:** Let $x, y \in \mathbb{N}$ such that $x \preceq y$ and $y \preceq x$. By definition, $y = xk$ and $x = ym$ for some $k, m \in \mathbb{N}$. Substituting x into the first equation yields $y = (ym)k = y(mk)$. Dividing both sides by y (since $y \geq 1$) gives $1 = mk$. Since m, k are positive integers, the only solution is $m = 1$ and $k = 1$. Substituting $k = 1$ into $y = xk$ gives $y = x$.
- **Transitive:** Let $x, y, z \in \mathbb{N}$ such that $x \preceq y$ and $y \preceq z$. By definition, $y = xk$ and $z = ym$ for some $k, m \in \mathbb{N}$. Substituting y into the equation for z gives $z = (xk)m = x(km)$. Since $k, m \in \mathbb{N}$, their product $km \in \mathbb{N}$. Thus, there exists an integer $k' = km$ such that $z = xk'$, meaning $x \mid z$, so $x \preceq z$.

Since the relation $\preceq$ is reflexive, antisymmetric, and transitive, it is a partial order on $\mathbb{N}$.

• Exercise 13

Inverse Relation

Let $\sim$ be a relation on a set X . The inverse relation $\sim^{-1}$ is defined by

$$x \sim^{-1} y \iff y \sim x.$$

Prove that $\sim$ is symmetric if and only if

$$\sim = \sim^{-1}.$$

◦ Solution 13.1

Proof:

($\implies$) Assume $\sim$ is symmetric. We prove $\sim = \sim^{-1}$ by showing mutual inclusion:

- Let $(x, y) \in \sim$, meaning $x \sim y$. Since $\sim$ is symmetric, we have $y \sim x$. By definition of the inverse relation, $y \sim x \iff x \sim^{-1} y$, which means $(x, y) \in \sim^{-1}$. Thus, $\sim \subseteq \sim^{-1}$.
- Let $(x, y) \in \sim^{-1}$, meaning $x \sim^{-1} y$. By definition of the inverse relation, this means $y \sim x$. Since $\sim$ is symmetric, $y \sim x \implies x \sim y$, which means $(x, y) \in \sim$. Thus, $\sim^{-1} \subseteq \sim$. Therefore, $\sim = \sim^{-1}$.

($\impliedby$) Assume $\sim = \sim^{-1}$. To show $\sim$ is symmetric, let $x, y \in X$ such that $x \sim y$, which means $(x, y) \in \sim$.

Since $\sim = \sim^{-1}$, it follows that $(x, y) \in \sim^{-1}$, meaning $x \sim^{-1} y$.

By definition of the inverse relation, $x \sim^{-1} y \iff y \sim x$.

Thus, $x \sim y \implies y \sim x$, proving that $\sim$ is symmetric.

• **Exercise 14**

Identity Relation and Antisymmetry

Let $\sim_X$ be the identity relation on X , defined by

$$x \sim_X y \iff x = y.$$

Let $\sim$ be a relation on X , and let $\sim^{-1}$ denote its inverse relation. Prove that $\sim$ is antisymmetric if and only if

$$\{(x, y) \in X \times X \mid x \sim y\} \cap \{(x, y) \in X \times X \mid x \sim^{-1} y\} \subseteq \{(x, y) \in X \times X \mid x \sim_X y\}.$$

◦ **Solution 14.1**

Proof:

For simplicity, let $R = \{(x, y) \in X \times X \mid x \sim y\}$, $R^{-1} = \{(x, y) \in X \times X \mid x \sim^{-1} y\}$, and $I = \{(x, y) \in X \times X \mid x \sim_X y\}$.

($\implies$) Assume $\sim$ is antisymmetric. Let $(x, y) \in R \cap R^{-1}$.

By definition of intersection, $(x, y) \in R$ and $(x, y) \in R^{-1}$.

This means $x \sim y$ and $x \sim^{-1} y$.

By definition of the inverse relation, $x \sim^{-1} y \iff y \sim x$.

Since $\sim$ is antisymmetric, $x \sim y \wedge y \sim x \implies x = y$.

By definition of the identity relation, $x = y \iff x \sim_X y \iff (x, y) \in I$.

Thus, $R \cap R^{-1} \subseteq I$.

($\impliedby$) Assume $R \cap R^{-1} \subseteq I$. To prove $\sim$ is antisymmetric, let $x, y \in X$ such that $x \sim y$ and $y \sim x$.

Since $x \sim y$, we have $(x, y) \in R$.

Since $y \sim x$, by definition of the inverse relation, we have $x \sim^{-1} y$, which means $(x, y) \in R^{-1}$.

Therefore, $(x, y) \in R \cap R^{-1}$.

By our inclusion hypothesis, $(x, y) \in R \cap R^{-1} \implies (x, y) \in I$.

By definition of I , $(x, y) \in I \iff x \sim_X y \iff x = y$.

Thus, $x \sim y \wedge y \sim x \implies x = y$, proving that $\sim$ is antisymmetric.

• **Exercise 15**

Which Relations Define Functions?

For each of the following relations $R \subseteq \mathbb{R} \times \mathbb{R}$, determine whether R defines a function

$$f: \mathbb{R} \rightarrow \mathbb{R}.$$

Explain your answer.

(a) $R = \{(x, y) \in \mathbb{R}^2 : y = x^2\}.$

(b) $R = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}.$

(c) $R = \{(x, y) \in \mathbb{R}^2 : xy = 1\}.$

(d) $R = \{(x, y) \in \mathbb{R}^2 : y^3 = x\}.$

◦ **Solution 15.1**

By definition, a relation $R \subseteq \mathbb{R} \times \mathbb{R}$ defines a function $f: \mathbb{R} \rightarrow \mathbb{R}$ if for every $x \in \mathbb{R}$, there exists a unique $y \in \mathbb{R}$ such that $(x, y) \in R$.

■ **(a) Yes.** For every $x \in \mathbb{R}$, the value $y = x^2$ is a uniquely determined real number.

■ **(b) No.** It fails on two conditions of the definition:

■ For $x = -1 \in \mathbb{R}$, there is no $y \in \mathbb{R}$ such that $y^2 = -1$.

■ For $x = 4 \in \mathbb{R}$, there is no unique y because both $y = 2$ and $y = -2$ satisfy $y^2 = 4$.

■ **(c) No.** For $x = 0 \in \mathbb{R}$, there is no $y \in \mathbb{R}$ such that $0 \cdot y = 1$. Thus, $f(0)$ is undefined, so it

cannot be a function from the entire domain $\mathbb{R}$.

- **(d) Yes.** For every $x \in \mathbb{R}$, every real number has a unique real cube root, so $y = \sqrt[3]{x}$ exists and is uniquely determined for all x .

• **Exercise 16**

Natural Domains and Ranges

Verify the natural domains and corresponding ranges of the following functions. The domain in each case consists of all values of x for which the formula is well-defined.

Function	Domain	Range
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

◦ **Solution 16.1**

- **For $y = x^2$:**
 - **Domain:** The expression x^2 is defined for all real numbers, so the domain is $(-\infty, \infty)$.
 - **Range:** Since the square of any real number is non-negative, $x^2 \geq 0$, and for any $y \geq 0$, $x = \sqrt{y}$ exists. Thus, the range is $[0, \infty)$.
- **For $y = \frac{1}{x}$:**
 - **Domain:** The expression is well-defined as long as the denominator is non-zero, so $x \neq 0$, which gives $(-\infty, 0) \cup (0, \infty)$.
 - **Range:** Since $y = \frac{1}{x}$, y can never be 0. For any $y \neq 0$, we can choose $x = \frac{1}{y}$, so the range is $(-\infty, 0) \cup (0, \infty)$.
- **For $y = \sqrt{x}$:**
 - **Domain:** The square root is defined for non-negative real numbers, so $x \geq 0$, giving the domain $[0, \infty)$.
 - **Range:** The principal square root function always yields non-negative values, so $y \geq 0$, and for any $y \geq 0$, $x = y^2$ is in the domain. Thus, the range is $[0, \infty)$.
- **For $y = \sqrt{4 - x}$:**
 - **Domain:** We require $4 - x \geq 0 \iff x \leq 4$, which gives the interval $(-\infty, 4]$.
 - **Range:** The square root outputs non-negative values, $y \geq 0$. For any $y \geq 0$, setting $y = \sqrt{4 - x} \iff y^2 = 4 - x \iff x = 4 - y^2$, which is always ≤ 4 . Thus, the range is $[0, \infty)$.
- **For $y = \sqrt{1 - x^2}$:**
 - **Domain:** We require $1 - x^2 \geq 0 \iff x^2 \leq 1 \iff -1 \leq x \leq 1$, which gives the interval $[-1, 1]$.
 - **Range:** Since $-1 \leq x \leq 1$, we have $0 \leq x^2 \leq 1$, which implies $0 \leq 1 - x^2 \leq 1$. Taking the square root gives $0 \leq y \leq 1$. For any $y \in [0, 1]$, $x = \sqrt{1 - y^2}$ is a valid domain element. Thus, the range is $[0, 1]$.

• **Exercise 17**

Domain, Codomain, and Image

Let

$$f: \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(n) = 2n.$$

Determine:

- (a) the domain of f ;
- (b) the codomain of f ;
- (c) the image of f .

Is the image equal to the codomain?

◦ **Solution 17.1**

■ **(a) Domain of f :**

By definition, for a function written as $f: A \rightarrow B$, the set A is the domain. Therefore, the domain of f is $\mathbb{Z}$ (the set of all integers).

■ **(b) Codomain of f :**

By definition, for a function written as $f: A \rightarrow B$, the set B is the codomain. Therefore, the codomain of f is $\mathbb{Z}$ (the set of all integers).

■ **(c) Image of f :**

The image of f is the set of all outputs generated by the function:

$$\text{Image}(f) = \{f(n) \mid n \in \mathbb{Z}\} = \{2n \mid n \in \mathbb{Z}\}$$

This is the set of all even integers, which can be written as $2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$.

■ **Is the image equal to the codomain?**

No. The image is the set of even integers $2\mathbb{Z}$, while the codomain is the set of all integers $\mathbb{Z}$. Since there are odd integers in the codomain (such as $1 \in \mathbb{Z}$) that do not belong to the image ($1 \notin 2\mathbb{Z}$), the image is a proper subset of the codomain, meaning they are not equal.

• **Exercise 18**

Identity and Inclusion

Let

$$A = \{1, 2, 3\}, \quad B = \{1, 2, 3, 4\}.$$

Define

$$\text{id}_A: A \rightarrow A, \quad \text{id}_A(x) = x,$$

and the inclusion map

$$\iota: A \rightarrow B, \quad \iota(x) = x.$$

- (a) Find all values of id_A and ι .
- (b) Do the two functions have the same rule?
- (c) Are they equal as functions? Explain.

◦ **Solution 18.1**

- **(a)** By evaluating each element of the domain $A = \{1, 2, 3\}$ under their respective definitions:

$$\text{id}_A(1) = 1, \quad \text{id}_A(2) = 2, \quad \text{id}_A(3) = 3$$

$$\iota(1) = 1, \quad \iota(2) = 2, \quad \iota(3) = 3$$

- **(b) Yes.** Both functions have the exact same assignment rule, which maps any input element x directly to itself without alteration.

- **(c) No.** By definition, two functions are equal if and only if their domains, codomains, and rules are all identical. Here, the codomain of id_A is $A = \{1, 2, 3\}$, whereas the codomain of ι is $B = \{1, 2, 3, 4\}$. Since their codomains are not the same set ($A \neq B$), they are not equal as functions.

• **Exercise 19**

Restriction of a Function

Let

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2 - 1.$$

Let

$$A = [-1, 1]$$

and define

$$g = f|_A.$$

- (a) Write the domain and codomain of g .
- (b) Find $g(-1)$, $g(0)$, $g(1)$.
- (c) Are f and g equal?
- (d) Do they have the same formula?

◦ **Solution 19.1**

- **(a)** By definition of a function restriction, restricting a function $f: X \rightarrow Y$ to a subset $A \subseteq X$ means the new function has domain A while retaining the original rule and codomain. Therefore:

- Domain of g is $A = [-1, 1]$.

- Codomain of g is $\mathbb{R}$.

- **(b)** Since $g(x) = f(x)$ for all $x \in A$, we substitute the values into the formula $x^2 - 1$:

$$g(-1) = (-1)^2 - 1 = 1 - 1 = 0$$

$$g(0) = (0)^2 - 1 = 0 - 1 = -1$$

$$g(1) = (1)^2 - 1 = 1 - 1 = 0$$

- **(c) No.** By definition, two functions are equal if and only if they share the exact same domain, codomain, and assignment rule. Here, the domain of f is $\mathbb{R}$ while the domain of g is $[-1, 1]$. Since their domains are different ($\mathbb{R} \neq [-1, 1]$), the functions are not equal.
- **(d) Yes.** Both functions use the exact same algebraic formula $x^2 - 1$ to compute outputs from inputs, although g is only allowed to apply this formula to elements within the restricted set A .

• **Exercise 20**

Construct Your Own Relation

Let

$$A = \{1, 2, 3, 4\}.$$

Construct:

- (a) a relation that is reflexive but not symmetric;
- (b) a relation that is symmetric but not reflexive;

- (c) an equivalence relation with exactly two equivalence classes;
- (d) a partial order different from the equality relation.

◦ **Solution 20.1**

- **(a)** To be reflexive, the relation must contain $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. To violate symmetry, we add a directional pair without its inverse, such as $(1, 2)$ without $(2, 1)$:

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2)\}.$$

- **(b)** To violate reflexivity, at least one diagonal element must be missing (e.g., $(1, 1) \notin R$). To maintain symmetry, any ordered pair must have its coordinates flipped within the set:

$$R = \{(1, 2), (2, 1)\}.$$

- **(c)** To form an equivalence relation with exactly two classes, we can partition A into two disjoint non-empty subsets, for instance, $A_1 = \{1, 2\}$ and $A_2 = \{3, 4\}$. The relation is constructed by taking $(A_1 \times A_1) \cup (A_2 \times A_2)$:

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}.$$

Explicitly rewritten for verification:

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}.$$

- **(d)** A partial order must be reflexive, antisymmetric, and transitive. The standard total order relation "less than or equal to" on integers satisfies this. Restricting it to A yields:

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

This relation is distinctly different from the pure equality relation because it contains pairs where $x \neq y$ (such as $(1, 2)$).

• **Exercise 21**

Let $f: \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = x^3 + 2$. Prove that f is bijective.

◦ **Proof 21.1**

- **Injectivity** Suppose $x, y \in \mathbb{R}$, suppose $f(x) = f(y)$, we want to prove that $x = y$. Then

$$f(x) = f(y) \iff x^3 + 2 = y^3 + 2$$

$$\iff x^3 = y^3$$

$$\iff x^3 - y^3 = 0$$

$$\iff (x - y)(x^2 + xy + y^2) = 0 = (x - y) \left(\left(x + \frac{y}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}y\right)^2 \right)$$

$$\iff x = y \vee \left(\left(x + \frac{y}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}y\right)^2 \right) = 0$$

$$\iff x = y \vee (x = 0 \wedge y = 0).$$

In any case $x = y$. Then f is injective.

- **Surjectivity** Suppose $y \in \mathbb{R}$. We want to prove $\exists x \in \mathbb{R}$ such that $x^3 + 2 = y \iff x^3 = y - 2$.

$$x := \sqrt[3]{y - 2} \implies f(x) = \left(\sqrt[3]{y - 2}\right)^3 + 2 = y - 2 + 2 = y.$$

Thus f is bijective.

• **Exercise 22**

Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$, defined by $f(n) = 3n + 1$. Is f injective? Is f surjective? Is f bijective?

◦ **Solution**

▪ **Injectivity** Suppose $n, m \in \mathbb{Z}$, $f(n) = f(m)$. By definition of f , $3n + 1 = 3m + 1 \implies 3n = 3m \implies n = m$.

▪ **Surjective** Suppose $y \in \mathbb{Z}$. We want to find $n \in \mathbb{Z}$ such that $f(n) = y = 3n + 1 \implies \frac{y-1}{3} = n$.

By contradiction. If $n \in \mathbb{Z}$ such that $f(n) = 3$ then $n = \frac{2}{3} \notin \mathbb{Z}$. Then there doesn't exist $n \in \mathbb{Z}$ such that $f(n) = 3$. Then $3 \notin \text{Im } f \implies f$ is not surjective. Thus f is not bijective.

• **Exercise 23**

Let $f: X \rightarrow Y$ be a function. Prove that f is injective $\iff$ For every set Z and for every pair of functions $g, h: Z \rightarrow X$, if $f \circ g = f \circ h \implies g = h$.

◦ **Proof 23.1**

Suppose $f: X \rightarrow Y$ a function.

($\implies$) Suppose f is injective. Suppose Z is a set, $g, h: Z \rightarrow X$ such that $f \circ g = f \circ h \implies \forall z \in Z, (f \circ g)(z) = (f \circ h)(z)$. Then $f(g(z)) = f(h(z))$. By the definition of injection, $g(z) = h(z)$ for all $z \in Z$ thus $g = h$.

($\impliedby$) Assume that for every set Z and for every pair of functions $g, h: Z \rightarrow X$, $f \circ g = f \circ h \implies g = h$ holds. We need to show that f is injective.

Let $x_1, x_2 \in X$ be arbitrary elements in the domain such that:

$$f(x_1) = f(x_2)$$

Since our hypothesis holds true for any choice of set Z , let us choose $Z = \{*\}$ to be a singleton set containing a single element.

We define two functions $g, h: \{*\} \rightarrow X$ by their assignments:

$$g(*) = x_1 \quad \text{and} \quad h(*) = x_2$$

Now, evaluate the compositions $f \circ g$ and $f \circ h$ on the single element $* \in Z$:

$$(f \circ g)(*) = f(g(*)) = f(x_1)$$

$$(f \circ h)(*) = f(h(*)) = f(x_2)$$

Since we assumed $f(x_1) = f(x_2)$, it follows that $(f \circ g)(*) = (f \circ h)(*)$. Since the domain contains only $*$, this implies:

$$f \circ g = f \circ h$$

By our hypothesis, this condition guarantees that $g = h$.

By definition of function equality, $g = h \implies g(*) = h(*)$.

Substituting our definitions back gives:

$$x_1 = x_2$$

Thus, $f(x_1) = f(x_2) \implies x_1 = x_2$ for any arbitrary elements, proving that f is injective.

The proof is complete.

• **Exercise 24**

Consider the functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{x}{1+|x|}, \quad g(x) = \begin{cases} \frac{x}{1+|x|} & (|x| < 1) \\ 0 & (|x| \geq 1) \end{cases}$$

1. Is g a left inverse of f ?

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) = g\left(\frac{x}{1+|x|}\right) \\ &= \frac{\frac{x}{1+|x|}}{1 + \left|\frac{x}{1+|x|}\right|} \\ &= \frac{\frac{x}{1+|x|}}{1 - \frac{|x|}{1+|x|}} \\ &= \frac{x}{1+|x|} \\ &= x \end{aligned}$$

Thus the statement is true.

2. Is g a right inverse of f ?

$$(f \circ g)(1) = f(g(1)) = f(0) = 0$$

Thus the statement is false

3. Is f invertible?

■ **Problem 28**

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ is **not invertible** because it is not surjective onto its given codomain $\mathbb{R}$.

■ **Analysis based on definitions:**

By definition, a function is invertible if and only if it is a bijection.

Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

- If $x_1, x_2 \geq 0$, then $\frac{x_1}{1+x_1} = \frac{x_2}{1+x_2} \implies x_1(1+x_2) = x_2(1+x_1) \implies x_1 = x_2$.
- If $x_1, x_2 < 0$, then $\frac{x_1}{1-x_1} = \frac{x_2}{1-x_2} \implies x_1(1-x_2) = x_2(1-x_1) \implies x_1 = x_2$.
- Since $f(x) \geq 0$ for $x \geq 0$ and $f(x) < 0$ for $x < 0$, inputs of different signs cannot have the same output. Thus, $f(x_1) = f(x_2) \implies x_1 = x_2$, showing that f is injective.

Let $y \in \mathbb{R}$ be an arbitrary element in the codomain. By observing the structure of the function, since $1 + |x| > |x|$ holds true for all $x \in \mathbb{R}$, we always have:

$$|f(x)| = \left| \frac{x}{1+|x|} \right| = \frac{|x|}{1+|x|} < 1$$

This means the image of f is bounded, specifically $\text{Im } f = (-1, 1)$. Because the image does not equal the given codomain ($\text{Im } f \neq \mathbb{R}$), there exist elements in the codomain for which no preimage $x \in \mathbb{R}$ can be found. Thus, f is not surjective.

• **Problem 25**

Determine the maximal possible domain D of $f(x) = \frac{x+1}{x+2}$ and the image $I = \text{Im } f$. Is $f: D \rightarrow I$ bijective, invertible? If so, find the inverse.

◦ **Solution 25.1**

- **Maximal Domain:** The rational function is well-defined as long as the denominator is non-zero, $x + 2 \neq 0 \iff x \neq -2$. Thus, the maximal possible domain is:

$$D = \mathbb{R} \setminus \{-2\}$$

- **Image:** By definition, $\text{Im } f = \{y \in \mathbb{R} \mid \exists x \in D, f(x) = y\}$. Let $y = \frac{x+1}{x+2}$. Rearranging the equation to solve for x :

$$y(x+2) = x+1 \implies yx + 2y = x+1 \implies x(y-1) = 1-2y \implies x = \frac{1-2y}{y-1}$$

The expression for x is well-defined for all $y \neq 1$. Furthermore, the value of x can never equal -2 because $\frac{1-2y}{y-1} = -2 \implies 1-2y = -2y+2 \implies 1=2$, which is impossible. Thus, the image is:

$$I = \mathbb{R} \setminus \{1\}$$

- **Bijectivity and Inversibility:**

- **Injective:** Let $x_1, x_2 \in D$ such that $f(x_1) = f(x_2)$. Then $\frac{x_1+1}{x_1+2} = \frac{x_2+1}{x_2+2} \implies (x_1+1)(x_2+2) = (x_2+1)(x_1+2) \implies x_1x_2 + 2x_1 + x_2 + 2 = x_1x_2 + 2x_2 + x_1 + 2 \implies x_1 = x_2$.

- **Surjective:** By restricting the codomain to the image I , every $y \in I$ has a preimage $x = \frac{1-2y}{y-1} \in D$ such that $f(x) = y$. Since $f: D \rightarrow I$ is both injective and surjective, it is bijective and therefore invertible.

- **Inverse Function:** The inverse function $f^{-1}: I \rightarrow D$ is given by:

$$f^{-1}(y) = \frac{1-2y}{y-1}$$

• **Problem 26**

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x^2}{x^2+1}$.

1. Find $f^{-1}\left(\frac{1}{2}\right)$.
2. Find $f^{-1}\left(\left[\frac{1}{4}, \frac{3}{4}\right]\right)$.
3. Let $A = [0, 3] \subset \mathbb{R}$, find $f(A)$, $f^{-1}(f(A))$.

◦ **Solution 26.1**

- **1.** By definition of a singleton preimage, we solve $f(x) = \frac{1}{2}$:

$$\frac{x^2}{x^2+1} = \frac{1}{2} \implies 2x^2 = x^2+1 \implies x^2 = 1 \implies x = \pm 1$$

Thus:

$$f^{-1}\left(\frac{1}{2}\right) = \{-1, 1\}$$

- **2.** By definition, $f^{-1}\left(\left[\frac{1}{4}, \frac{3}{4}\right]\right) = \left\{x \in \mathbb{R} \mid \frac{1}{4} \leq \frac{x^2}{x^2+1} \leq \frac{3}{4}\right\}$. Since $x^2+1 > 0$, we solve the inequalities separately:

- Left side: $\frac{1}{4}(x^2+1) \leq x^2 \implies x^2+1 \leq 4x^2 \implies 3x^2 \geq 1 \implies x^2 \geq \frac{1}{3} \implies x \in \left(-\infty, -\frac{1}{\sqrt{3}}\right] \cup \left[\frac{1}{\sqrt{3}}, \infty\right)$.

- Right side: $x^2 \leq \frac{3}{4}(x^2+1) \implies 4x^2 \leq 3x^2+3 \implies x^2 \leq 3 \implies x \in [-\sqrt{3}, \sqrt{3}]$.

$$\in [-\sqrt{3}, \sqrt{3}].$$

Intersecting the two solution intervals yields:

$$f^{-1}\left(\left[\frac{1}{4}, \frac{3}{4}\right]\right) = \left[-\sqrt{3}, -\frac{1}{\sqrt{3}}\right] \cup \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$$

- 3. Note that the question text has a standard typo writing $A \subset \mathbb{R}^3$, but the interval $[0, 3]$ indicates $A \subset \mathbb{R}$.
 - To find $f(A)$, since $f(x) = \frac{x^2}{x^2+1} = 1 - \frac{1}{x^2+1}$, the function is strictly increasing on $[0, 3]$. Evaluating at the endpoints gives $f(0) = 0$ and $f(3) = \frac{9}{10}$. Thus:

$$f(A) = \left[0, \frac{9}{10}\right]$$

- To find $f^{-1}(f(A))$, we solve $x \in \mathbb{R}$ such that $0 \leq \frac{x^2}{x^2+1} \leq \frac{9}{10}$. The left inequality is true for all $x \in \mathbb{R}$. The right inequality gives $10x^2 \leq 9x^2 + 9 \implies x^2 \leq 9 \implies -3 \leq x \leq 3$. Thus:

$$f^{-1}(f(A)) = [-3, 3]$$

• **Problem 27**

Let $f: X \rightarrow Y$ be a function. Prove that f is injective $\iff \forall A \subseteq X, f^{-1}(f(A)) \subseteq A$.

◦ **Solution 27.1**

Proof:

($\implies$) Assume f is injective. Let $A \subseteq X$ be an arbitrary subset, and let $x \in f^{-1}(f(A))$.

By definition of preimage, $x \in f^{-1}(f(A)) \implies f(x) \in f(A)$.

By definition of image, $f(x) \in f(A) \implies \exists x' \in A$ such that $f(x) = f(x')$.

Since $x, x' \in X$ and f is given to be injective, $f(x) = f(x') \implies x = x'$.

Since $x' \in A$, it follows directly that $x \in A$.

Thus, $f^{-1}(f(A)) \subseteq A$.

($\impliedby$) Assume that for every subset $A \subseteq X$, $f^{-1}(f(A)) \subseteq A$ holds true. We need to show that f is injective.

Let $x_1, x_2 \in X$ be arbitrary elements in the domain such that:

$$f(x_1) = f(x_2)$$

Since our hypothesis holds true for *any* subset of X , let us choose $A = \{x_1\}$ to be the singleton subset containing only x_1 .

Evaluate the image of A : $f(A) = \{f(x_1)\}$.

Since we assumed $f(x_1) = f(x_2)$, it follows that $f(x_2) \in \{f(x_1)\} = f(A)$.

By definition of preimage, $f(x_2) \in f(A) \implies x_2 \in f^{-1}(f(A))$.

By our hypothesis, $f^{-1}(f(A)) \subseteq A$, which means $x_2 \in A$.

Since $A = \{x_1\}$, the statement $x_2 \in \{x_1\}$ implies:

$$x_2 = x_1$$

Thus, $f(x_1) = f(x_2) \implies x_1 = x_2$, proving that f is injective.

The proof is complete.